

PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

F. Min Path

time limit per test: 1 second
memory limit per test: 256 megabytes

You are given a graph with n vertices (numbered 1 through n) and m **directed weighted** edges. Find the minimum possible total weight of a path with k edges. A path can visit the same vertex or edge multiple times.

If there is no path with k edges, print IMPOSSIBLE instead.

Input

The first line contains three integers n , m and k ($1 \leq n \leq 100$, $0 \leq m \leq n \cdot (n - 1)$, $1 \leq k \leq 10^9$) — the number of vertices, the number of edges and the length of a path.

The following m lines describe edges. The i -th of these lines contains three integers a_i , b_i and c_i ($1 \leq a_i, b_i \leq n$, $a_i \neq b_i$, $-10^9 \leq c_i \leq 10^9$), denoting a directed edge from a_i to b_i of weight c_i . There are no self-loops or multiple edges.

Output

Print the minimum possible sum of weights of a path with k edges, or the word IMPOSSIBLE if there is no such path.

Examples

input	Copy
<pre>3 4 2 1 2 12 2 3 -5 3 1 20 2 1 -1</pre>	
output	Copy
<pre>7</pre>	

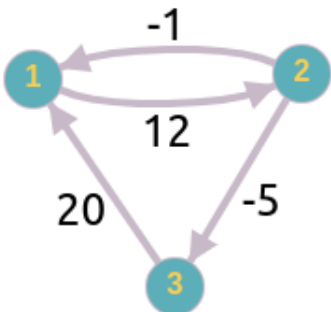
input	Copy
<pre>3 4 5 1 2 12 2 3 -5 3 1 20 2 1 -1</pre>	
output	Copy
<pre>17</pre>	

input	Copy
<pre>6 5 3 1 3 -50 2 4 -51 4 3 -52 3 6 -53 4 5 -54</pre>	
output	Copy
<pre>-156</pre>	

input	Copy
<pre>6 5 4 1 3 -50 2 4 -51 4 3 -52 3 6 -53 4 5 -54</pre>	
output	Copy
<pre>IMPOSSIBLE</pre>	

Note

The drawing below shows the graph in two first sample tests. The optimal paths are respectively $1 \rightarrow 2 \rightarrow 3$ and $2 \rightarrow 1 \rightarrow 2 \rightarrow 1 \rightarrow 2 \rightarrow 3$.



Matrix Exponentiation

Finished

Practice

→ Virtual participation

▼

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

▼

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language:

GNU G++23 14.2 (64 bit, ms▼

Choose file:

Choose File

No file chosen

Submit

→ Last submissions		
Submission	Time	Verdict
336306816	Aug/30/2025 22:32	Happy New Year!
336306468	Aug/30/2025 22:28	Delivered to wrong address on flight 4 WA4